PyPath

Python Ecopath with Ecosim

User Manual & Reference Guide

Version 0.5.0 | October 2026

PyPath is a comprehensive Python implementation of the Ecopath with Ecosim (EwE) ecosystem modeling framework.

1. Getting Started

What is PyPath?

PyPath is a Python implementation of Ecopath with Ecosim (EwE), the most widely used ecosystem modeling framework in fisheries science.

2. Installation

Using Conda

```
conda install pypath-ewe pypath-shiny -c razinka -c conda-forge
```

Using Pip

```
pip install pypath-ewe pypath-shiny
```

Verify

```
python -c "import pypath; print(pypath.__version__)"
```

3. Quick Start

Web Application

```
pypath-shiny
```

Opens at http://localhost:8000. Use guided workflow to import data, balance model, and run simulations.

Python Library

```
from pypath.core.ecopath import rpath
from pypath.core.params import create_rpath_params

params = create_rpath_params(groups=['Predator', 'Prey'],
                             ee=[0.05, 0.1])
model = rpath(params)
```

4. Core Concepts

EwE Framework

ECOPATH: Static mass-balance model
ECOSIM: Dynamic time simulation
ECOSPACE: Spatial modeling

Functional Groups

Groups represent organisms: Similar size, diet, habitat, ecosystem role. Parameters: Biomass (B), Production/Biomass (P/B), Consumption/Biomass (Q/B), Ecotrophic Efficiency (EE)

5. Ecopath Model

Creating Model

Requires: Biomass estimates, Production/consumption rates, Diet matrix, Fishing data, Ecotrophic efficiency

Data Sources

Field surveys, Stock models, FISHBASE/SEALIFEBASE, Bioenergetics, Stomach analysis

6. Ecosim Simulation

Overview

Extends Ecopath with time dimension. Models fishing pressure, environmental forcing, ecosystem responses. Uses ODE integration with monthly steps.

Running

```
from pypath.core.ecosim import rsim_scenario, rsim_run
scenario = rsim_scenario(model, params)
results = rsim_run(scenario)
```

7. Ecospace Spatial

Features

Models ecosystem across habitat patches. Includes species dispersal, spatial fishing, habitat preferences. Useful for marine protected area design.

8. Individual-Based Model

IBM Overview

PyPath includes complete lifecycle IBM for Baltic smelt. Models eggs, yolk-sac larvae, juveniles, and adults with explicit ontogenetic processes.

9. Shiny Web App

Workflow

1. Data import (CSV, EwE)
2. Ecopath - Build and balance
3. Pre-balance checks
4. Ecosim - Configure simulation
5. Analysis - Visualize
6. Export results

10. API Reference

Main Imports

```
from pypath.core.ecopath import rpath
from pypath.core.ecosim import rsim_run, rsim_scenario
from pypath.core.params import create_rpath_params
from pypath.io.ewemdb import read_ewedb
```

Classes

RpathParams: Model parameters container. Rpath: Mass-balance result. RsimScenario: Simulation config. RsimOutput: Results.

11. Examples

Simple Model

```
params = create_rpath_params(
    groups=['Predator', 'Forage', 'Zoo'],
    ee=[0.05, 0.1, 0.2])
model = rpath(params)
```

Import from EwE

```
from pypath.io.ewemdb import read_ewedb
params = read_ewedb('model.eweaccdb')
model = rpath(params)
```

12. Troubleshooting

Issues

Model won't balance: Check diet sums to 1, verify EE (0-1), ensure parameters for all groups

Disconnect: Check network, review browser console, reload page

Bad results: Validate Ecopath, check fishing values, review parameters

13. FAQ

Compatible with EwE? Yes, import .eweaccdb files
For publication? Yes, peer-reviewed software
Commercial? Yes, MIT licensed
Report bugs? GitHub: https://github.com/razinkele/PyPath/issues

Resources

API: https://razinkele.github.io/PyPath/
GitHub: https://github.com/razinkele/PyPath
Demo: https://laguna.ku.lt/pypath/